

Preventive Maintenance Master Schedule

Daily:

- Verify proper inlet gas pressure.
- Verify proper gas flow settings. – Mandatory at every consumable change.
- Inspect torch and replace consumables as needed.

Weekly:

Week	Jan	Feb	March	April	May	June	July	Aug	Sept	Oct	Nov	Dec
1												
2												
3												
4												
5												

- Clean the power supply with dry, oil-free compressed air, or a vacuum cleaner.
- Verify cooling fans are working properly.
- Clean torch threads and current ring.
- Verify proper coolant level.

Semi-Annually:

Year	1 st Service	2 nd Service

- Replace service parts per the Service Part Replacement Schedule.

Annually:

Year									
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- Replace service parts per the Service Part Replacement Schedule.

Preventive Maintenance Protocol Checklist

Customer: _____
Location: _____
Contact: _____
Date: _____

Hypertherm system: _____
System serial #: _____
System arc hours: _____
(if equipped with an hour meter)

Comments *P* – Performed *NP* – Not present on system

Power supply

- ☐ *P* ☐ *NP* 1. Inspect for particulates and blow out
- ☐ *P* ☐ *NP* 2. Inspect wiring harnesses
- ☐ *P* ☐ *NP* 3. Inspect main contactor
- ☐ *P* ☐ *NP* 4. Inspect pilot arc relay

Coolant system

- ☐ *P* ☐ *NP* 5. Inspect filter element
- ☐ *P* ☐ *NP* 6. Perform coolant flow test
 - A. Coolant flow checked at _____
gallons per minute (_____ l/min)

Torch main body

- ☐ *P* ☐ *NP* 7. Inspect water tube
- ☐ *P* ☐ *NP* 8. Inspect current ring
- ☐ *P* ☐ *NP* 9. Clean threads on torch front end
- ☐ *P* ☐ *NP* 10. Inspect Vespel torch insulator
- ☐ *P* ☐ *NP* 11. Inspect torch and consumable o-rings
- ☐ *P* ☐ *NP* 12. Verify proper fit of retaining or shield cap
- ☐ *P* ☐ *NP* 13. Inspect hose fittings
- ☐ *P* ☐ *NP* 14. Inspect torch-to-torch-lead connections

Gas flows

- ☐ *P* ☐ *NP* 15. Inspect plumbing from gas supply
 - ☐ *P* ☐ *NP* A. Oxygen
 - ☐ *P* ☐ *NP* B. Nitrogen
 - ☐ *P* ☐ *NP* C. Air
 - ☐ *P* ☐ *NP* D. Nitrogen-Hydrogen
 - ☐ *P* ☐ *NP* E. Argon-Hydrogen
 - ☐ *P* ☐ *NP* F. Inspect compressed air filter system

Gas flows – continued

- ☐ *P* ☐ *NP* 16. Perform gas leak tests
 - A. Oxygen pressure drop at _____
psi in 10 minutes (_____ bar)
 - B. Nitrogen pressure drop at _____
psi in 10 minutes (_____ bar)
- ☐ *P* ☐ *NP* 17. Inspect for hose restrictions
 - ☐ *P* ☐ *NP* A. Metering console hoses
 - ☐ *P* ☐ *NP* B. Selection console to Metering console
 - ☐ *P* ☐ *NP* C. Metering console to torch
 - ☐ *P* ☐ *NP* D. Hoses in power track

Cable connections

- ☐ *P* ☐ *NP* 18. Inspect cables
 - ☐ *P* ☐ *NP* A. CommandTHC
 - ☐ *P* ☐ *NP* B. Control cable from metering console to selection console
 - ☐ *P* ☐ *NP* C. From metering console to power supply
 - ☐ *P* ☐ *NP* D. From ignition console and selection console to power supply

Ignition console

- ☐ *P* ☐ *NP* 19. Inspect for moisture, dust and particulates
- ☐ *P* ☐ *NP* 20. Inspect spark gap subassembly
- ☐ *P* ☐ *NP* 21. Inspect torch leads

System grounding

- ☐ *P* ☐ *NP* 22. Inspect for proper system component grounding
- ☐ *P* ☐ *NP* 23. Inspect connection from cutting table to workpiece (+) lead

General comments and recommendations:

Preventive maintenance performed by: _____ Date: _____

Service parts replacement schedule – 1 of 2

228016 – annual preventive maintenance parts kit

This kit includes all the parts in the shaded boxes below

Timeline	Component	Part number	Qty.
6 months or 300 arc hours	Coolant filter element	027664	1
	Coolant solution 70/30	028872	4
	Torch kit	128879	1
	Quick-disconnect kit	128880	1
1 Year or 600 arc hours	Coolant filter element	027664	1
	Coolant solution 70/30	028872	4
	Main contactor	003217	1
	Torch main body	220162	1
	Pilot arc relay	003149	1
1.5 Years or 900 arc hours+	Coolant filter element	027664	1
	Coolant solution 70/30	028872	4
	Torch kit	128879	1
	Quick-disconnect kit	128880	1
2 Years or 1200 arc hours	Coolant filter element	027664	1
	Coolant solution 70/30	028872	4
	Main contactor	003217	1
	Torch main body	220162	1
	Pilot arc relay	003149	1
	Coolant pump kit	228171	1
	Torch leads	System dependent	1
2.5 Years or 1500 arc hours	Coolant filter element	027664	1
	Coolant solution 70/30	028872	4
	Torch kit	128879	1
	Quick-disconnect kit	128880	1
3 Years or 1800 arc hours	Coolant filter element	027664	1
	Coolant solution 70/30	028872	4
	Main contactor	003217	1
	Torch main body	220162	1
	Pilot arc relay	003149	1
	Cooling fan 6"	127039	4
	Cooling fan 10"	027079	3
3.5 Years or 2100 arc hours	Coolant filter element	027664	1
	Coolant solution 70/30	028872	4
	Torch kit	128879	1
	Quick-disconnect kit	128880	1
4 Years or 2400 arc hours	Coolant filter element	027664	1
	Coolant solution 70/30	028872	4
	Main contactor	003217	1
	Torch main body	220162	1
	Pilot arc relay	003149	1
	Coolant pump kit	228171	1
	Torch leads	System dependent	1
	Coolant pump motor kit	128385	1
4.5 Years or 2700 arc hours	Coolant filter element	027664	1
	Coolant solution 70/30	028872	4
	Torch kit	128879	1
	Quick-disconnect kit	128880	1

Service parts replacement schedule – 2 of 2

Timeline	Component	Part number	Qty.
5 Years or 3000 arc hours	Coolant filter element	027664	1
	Coolant solution 70/30	028872	4
	Main contactor	003217	1
	Torch main body	220162	1
	Pilot arc relay	003149	1
	High voltage transformer	129854	1
	Power distribution board	041802	1
	Metering console	078184	1
	Pilot arc lead	System dependent	1
	Gas leads	System dependent	1
5.5 Years or 3300 arc hours	Coolant filter element	027664	1
	Coolant solution 70/30	028872	4
	Torch kit	128879	1
	Quick-disconnect kit	128880	1
6 Years or 3600 arc hours	Coolant filter element	027664	1
	Coolant solution 70/30	028872	4
	Main contactor	003217	1
	Torch main body	220162	1
	Pilot arc relay	003149	1
	Coolant pump kit	228171	1
	Torch leads	System dependent	1
	Cooling fan 6"	127039	4
	Cooling fan 10"	027079	3
6.5 Years or 3900 arc hours	Repeat schedule starting at 6 months or 300 arc hours		